Abstract

Overall scientific questions.

Introduction

- Overview of the ISM. How dust forms, why we use CO as a tracer. Why we might use other tracers.
- Jeans collapse a model for star formation.
- Quasistatic turbulent core fragmentation as a model for star formation.
- IMF / CMF. How the high mass end changes. SF across different environments.
- (Nico and Alessio) Literature on decoupling cores from clumps. We would like to know at what stage the gas decouples.
- CHIMPS paper and other extragalactic attempts at this sort of work.

The interstellar medium (ISM) is made up of primarily gas, as well as dust and cosmic rays. The ISM can be divided up based on temperature, ionisation and density. It consists of 5 main phases: the cold molecular medium (CMM), which is defined such that the gas is molecular with a temperature range of $10 - 30\ K$ and a number density between $10^2 - 10^6\ cm^{-3}$; the cold neutral medium (CNM), which is defined such that the gas is mostly atomic with a temperature range of $30 - 100\ K$ and a number density between $1 - 100\ cm^{-3}$; the warm neutral medium (WNM), which is defined such that the gas is mostly atomic with a temperature range of $6000 - 10000\ K$ and number densities below $1\ cm^{-3}$; the warm ionised medium (WIM), which is defined such that the gas is mostly ionised with a temperature range of $8000 - 10000\ K$ and number densities around $0.4\ cm^{-3}$ and the hot ionised medium (HIM), which is defined such that the gas is mostly ionised with a temperature range of $10^6 - 10^7\ K$ and number densities below $10^{-2}\ cm^{-3}$ (see Klessen and Glover 2016). The focus of this work will be on the CMM.

Other than primordial elements, every atom formed inside a star. Therefore, initially, these elements that make up the gas are hot and have not formed in their molecular state. To get to the temperature of the CMM, the gas must be cooled. A star ejects matter throughout its life as stellar winds, as well as at the end of its life, in different ways depending on its mass. As the gas moves away from its progenitor, the number of ionising photons will decrease which will offset the balance between ionisation and recombination. At these higher temperatures, the cooling from Lyman-α becomes important. The general form of the cooling rate of a two-level system is shown below.

$$\Lambda_{ul} = \frac{g_u}{g_l} A_{ul} E_{ul} n_l e^{-E_{ul}/k_B T} \tag{1}$$

Equation 1 is the rearranged form of 2.38 in Tielens 2005. The formula comes about by assuming LTE and that the number density is much less than the critical density. g_u and g_l are statistical weights of the energy states, A_{ul} is the Einstein A coefficient of spontaneous emission, E_{ul} is the energy of the transition, and n_l is the number density of the lower state.

$$\Lambda_{HI} = 7.3 \times 10^{-19} e^{-118400/T} n_e n_H \tag{2}$$

Equation 2 appears as 2.70 in Tielens 2005. Above $10^4 K$, the exponential term stops being negligible, which causes the cooling rate to rise rapidly. This together with recombination to form atomic hydrogen is what facilitates the transition to the WNM. At higher metallicities the cooling is not dominated by Lyman-α but by other fine structure transitions from heavier elements. Once cooled, Equation 2 shows a ceiling for the temperature in the WNM above which the cooling rate can rapidly exceed the different heating rates (Tielens and Hollenbach 1985). A different fine structure transition dominates the cooling in the WNM and facilitates the transition to the CNM, the CII transition. The energy of this transition is $92K$ (Hollenbach and McKee 1989), meaning the exponent in Equation 1 becomes $e^{-92/T}$, allowing the gas to cool efficiently down to $\sim 100K$. The term in the exponent implies that below $100K$ the cooling from CII decreases rapidly. On the other hand, the first rovibrational transition ($J = 1$–0) of CO emits a photon with a rest frequency of $115.27GHz$ (Winnewisser et al. 1997), which gives an energy of $5.53K$, using $h\nu/k_B$. This allows the cooling rate of CO to dominate in this temperature regime and allow the gas to get to a temperature of around $10K$. Even though CO is a more efficient coolant, CII seems to be a more dominant cooling mechanism at low metallicity as well as for diffuse gas with densities less than $10^3 cm^{-3}$ (Glover and Clark 2012). This is partly because at low metallicity, the CO cannot be shielded by the dust and photodissociates by way of the ISRF. It is also important to note that Glover and Clark 2012 indicate that if the cloud is shielded from the effects of photoelectric heating then cooling from CII can allow the gas to get down to temperatures of $20K$ without the presence of CO.

Once there is gas at the typical temperatures of the CMM, molecular hydrogen can begin to form. Molecular hydrogen should form directly between two atoms of hydrogen but the Einstein A coefficient is too small for the process to allow for a photon to be emitted during their collision (Wakelam et al. 2017). This means that molecular hydrogen cannot form efficiently this way, therefore another mechanism is required to get the amount of molecular hydrogen expected. The formation process involves atomic hydrogen sticking to a dust grain and colliding with another stuck atom of hydrogen allowing for the recombination of molecular hydrogen. The energy produced by this is enough to break the bonds between the dust grain and the molecule of hydrogen, allowing for it to return to the CMM (Wakelam et al. 2017). This has been confirmed experimentally using various dust grain substitutes made of carbon or silicates. Katz et al. 1999 find that the peak efficiency of the recombination occurs below $20K$ for both types of grain. The formation of CO is very complex (see Glover et al. 2010) and can involve different pathways, the simplest of which is the formation from a carbon atom and a hydroxyl radical. Even though it is thought that the gas being in a molecular state is not necessary for star formation, the cloud being able to shield itself from the ISRF is paramount to the formation of both stars and molecular gas (Glover and Clark 2012). They find that a cloud of atomic gas is still capable of forming stars at the same rate as molecular gas is able to form. This however does not change the fact that molecular hydrogen still traces star formation, it just may not trace all star formation.

Tracing molecular gas is important because it allows for the understanding of the steps that lead to star formation. CO is regarded as a good tracer of molecular gas for its abundance in molecular clouds and spontaneous low energy transitions, like the one described above. The idea is to trace the more abundant molecular hydrogen, which is not directly possible due to its lack of emission at the typical temperatures of the CMM. The lowest possible rotational transition of molecular hydrogen is at an energy of $510K$ (Wakelam et al. 2017). CO has been the gold standard when it comes to tracing molecular hydrogen, but tracers like HCN or N_2H^+ have been found to trace the dense gas much more effectively than CO(Kauffmann et al. 2017). N_2H^+ is almost always being detected above column densities of $10^{22} cm^{-2}$ (Fehér et al. 2025), meaning it is much better at tracing the clumps where stars are likely to be forming.

Regions of the ISM that contain the CMM are also known as molecular clouds. These clouds are the densest constituents of the ISM, the collapse of which leads to star formation. For a cloud to collapse the gravitational force pulling the gas together must overcome that of the internal pressure of the gas. Hydrostatic equilibrium is the point at which both forces cancel one another out. This allows for the derivation of the mass above which a spherical system would collapse, known as the Jeans mass.

$$M_J = \frac{\pi^{3/2} {c_s}^3}{{\rho_o}^{1/2} G^{3/2}} \quad (3)$$

Equation 3 appears as the rearranged form of 9.23 in Stahler and Palla 2004 using the fact that $M_J \sim \rho_o \lambda^3$. c_s refers to the isothermal sound speed in the medium given by $\sqrt{k_B T/\mu m_p}$. This equation assumes that the system is isolated with no external forces, but the ISM is turbulent therefore an external pressure could be present. In this instance the external pressure would oppose the internal pressure of the gas, making it easier for the gas to collapse. This mass is known as the Bonner-Ebert mass.

$$M_{BE} = \frac{1.18 {c_s}^4}{{P_o}^{1/2} G^{3/2}} \quad (4)$$

Equation 4 appears as 9.16 in Stahler and Palla 2004. P_o is the external pressure on the cloud. There are several mechanisms that affect star formation that these equations do not account for. Magnetic fields oppose the gravitational collapse of clouds, slowing or stopping the collapse of clouds (Krasnopolsky and Gammie 2005). This allows more gas to be accreted before collapse leading to higher mass star formation. Stellar feedback strips the diffuse gas from the cloud slowing their growth but allowing the dense centres to continue to be star forming (Watkins et al. 2025). It can also cause the destruction of the host cloud. Watkins et al. 2025 also found that stellar feedback likely breaks apart larger clouds into smaller clouds. Protostellar radiative feedback stops fragment formation causing most of the gas to be accreted onto a few objects leading to higher mass star formation (Krumholz, Klein, and McKee 2007). Cloud-cloud collisions between both atomic and molecular clouds facilitates the formation of massive molecular cloud cores, making them the dominant mechanism for forming high mass stars. Additionally, there is observational evidence that they are responsible for starbursts (e.g. the Antenna galaxies) (Fukui et al. 2021).

As a cloud collapses, the Jeans mass scales proportionally with temperature and density $\left(M_J \propto \rho^{-1/2}T^{3/2}\right)$, as shown by Equation 3. During gravitational collapse the density always increases and providing the gas is isothermal, the Jeans mass decreases as the density increases (Ward-Thompson and Whitworth 2011). Therefore, sections of the collapsing cloud become Jeans unstable and begin gravitational collapse causing multiple collapsing regions. These collapsing regions can go on to fragment again. Eventually the density increases so much so heat is unable to radiate away and cooling becomes unable to keep the gas isothermal, and so temperature increases. Assuming the collapsing cloud acts like a polytrope, it will follow the equation of state ($P = K\rho^{\gamma}$), where γ is the adiabatic index (Chieze 1987). If heat cannot radiate away then the collapse can be considered adiabatic, so the temperature is now a function of density $\left(T = K\rho^{\gamma-1}\right)$. This now means that the Jeans mass only scales proportionally with density $\left(M_J \propto \rho^{(3/2)(\gamma-4/3)}\right)$. In molecular clouds, the gas is primarily diatomic or monatomic which have 5 and 3 degrees of freedom, which gives them adiabatic indices of 5/3 and 7/5, respectively ($\gamma = 1 + 2/d.o.f.$). Both these indices result in a direct proportionally between the Jeans mass and density, causing fragmentation to cease.

The above picture of fragmentation is not complete as it does not take into account turbulence or magnetic fields. Magnetic fields as described above stop fragmentation by exerting a force on charged particles outwards. The charged particles then collide with neutral matter causing their velocities to couple (Krasnopolsky and Gammie 2005). It is for this reason that ambipolar diffusion was introduced, which allows the star forming material to be decoupled from the ions affected by strong magnetic fields. This occurs when the clouds get sufficiently dense that cosmic rays and the ISRF are unable to cause substantial ionisation, which reduces the rate at which the collisions occur (Mestel and Spitzer 1956). This allows the charged particles to pass through the neutral matter without imposing its velocity, allowing the neutral matter to go on to fragment and form stars.

Ambipolar diffusion is a slow process and is unable to reproduce the numerical formation time of a core from observations ($\sim 1Myr$) (Ciolek and Basu 2001). This can be solved with the addition of supersonic turbulence (Mac Low and Klessen 2004). The dominant driving process of this turbulence in star forming galaxies is supernovae, which self regulates by temporarily suppressing star formation, leading to high mass star formation and thus more supernovae. These high mass stars during their lifetimes produce ionising radiation that contributes short lasting turbulence by heating the gas which then contracts as it cools (Kritsuk and Norman 2002). Gravitational instabilities are a local driving mechanism of the turbulence but are unable to delay collapse for much more than a free-fall time due to the fact that the turbulence decays. Magnetorotational instabilities allow for turbulent flows to be generated from shear by magnetic fields generating Maxwell stress, which may occur at the outskirts of the Milky Way disc. The supersonic turbulence generated by these driving forces builds up gas at its shock front, forming dense sheets and filaments. These dense regions even form in the presence of magnetic fields. These driving forces produce random motions in the gas that can act against gravitational collapse temporarily slowing the infall. This supersonic turbulence can only do so as long as it is continuously driven because it decays quickly. For an isothermal shock the density scales with the Mach number $\left(\rho' = \mathcal{M}^2\rho\right)$, where the Mach number is directly proportional to the rms velocity dispersion due to turbulence. If the turbulence is treated as an additional pressure then the effective sound speed can be defined which takes into account the rms velocity dispersion ($c_{\mathrm{s,eff}}^2 = c_s^2 + \nu_{\mathrm{rms}}^2/3$) (Chandrasekhar 1951). It is assumed that $\nu_{\mathrm{rms}} >> c_s$ because the turbulence is supersonic, which means that the Jeans mass now is a function of the turbulence as well as the density $\left(M_J \propto \nu_{\rm rms}^3 \rho^{-1/2}\right)$. This means that an increase in turbulence increases the Jeans mass, preventing collapse. Eventually the turbulence decays and the cloud can collapse. If the clump is able to become sufficiently dense that it decouples from the ambient gas in the cloud, a second shock will be unable to disrupt it. If the gas is not able to collapse in this time then a second shock would disrupt the clump inducing further turbulence and preventing collapse further. These successive shocks in a short amount of time is the basis for high mass star formation in this theory (see Mac Low and Klessen 2004).

IMF

Project Work

- Motivations (link back to the lit review).
- Methods.
- Conclusions and what that might mean.
- Future work.

It is aparanent from the introduction that different stellar processes have a direct impact on star formation and affect its rate as well as the mass of the stars forming. Some of these stellar processes can cause changes to the kinematics of the molecular clouds. Before any change in the kinematics due to these processes, the gas is expected to follow the flow imparted by the galaxy during its formation. At what stage during star formation do the dynamics of the galaxy no longer dominate, and the effects of one or more of the processes described above are being seen? The aim of the following work is to answer this question.

The obvious place to conduct this research is the Milky Way due to its proximity, meaning high resolution surveys can be conducted on the molecular clouds. To probe different levels of molecular clouds, different density tracers, which are only present at certain stages of star formation, can be used. ^{12}CO is thought to be present at all stages but due to its high abundance, it is optically thick in the denser regions making a good tracer fo the more diffuse gas at the edges of molecular clouds. ^{13}CO, as an isotope, is less abundant, making it better at tracing the denser material in molecular clouds. It shows the extent of the entire cloud, which is the main reason why it is used to estimate the column density. HCN traces the intermediately dense gas in the cloud. N_2H^+ traces the truly dense gas in clumps that is actively collapsing.

The Structure, Excitation and Dynamics of the Inner Galactic Interstellar Medium (SEDIGISM) survey (Schuller et al. 2021) traces the ($J = 2$–1) transition of ^{13}CO, as well as several other tracers not discussed here. This project starts with making use of this survey. The SEDIGISM survey covers at least 0.5 degrees in galactic latitude either side of the galactic plane and 77 degrees in galactic longitude across mainly the fourth quadrant and part of the first quadrant ($300° \leq \ell \leq 18°$). The observations are spectral cubes where the spectral axes, which contains 1600 channels, have been converted into velocity in kms^{-1}. An individual slice of the spectral cube is typically a size of $\sim 775 \times 398$ voxels. A clustering algorithm was developed to extract the clouds from the survey by using graph theory to seperate emission lines, this algorithm is called Spectral Clustering for Interstellar Molecular Emission Segmentation (SCIMES) (Colombo et al. 2015). The cloud extraction yeileded 10663 clouds after dealing with overlapping areas (Duarte-Cabral et al. 2021). A final catalogue was produced as well as masks, which contained zeroes everywhere other than the voxels which successfully made it through the dendrogram. Those that do not conatin zeroes contain the cloud ID.

For every spatial coordinate, the masks give the voxels that contain significant ^{13}CO(2–1) emission. Therefore, information about the spectrum can be obtained from the voxels in the survey that the masks correspond to. In spectroscopy, there exists different moments which show different aspects of a spectrum. A piece of code was written that calculated the zeroth, first and second moment for all clouds in the survey, creating output FITS files containing the data for each moment.

$$M_0 = \Delta\nu \sum_{i=0}^{n} I_i \tag{5}$$

Equation 5 shows the equation for the zeroth moment of the velocity or the integrated flux. $\Delta\nu$ represents the channel width in kms^{-1}, I_i represents the flux in an individual voxel. The flux is summed for every voxel for an individual spatial coordinate. Equation 5 gives the area under the emission line, which is the total flux of the ^{13}CO for each spatial coordinate. All the moment equations are calculated once for each spatial coordinate in each cloud.

$$M_1 = \frac{\Delta\nu \sum_{i=0}^{n} I_i \nu_i}{M_0} \tag{6}$$

Equation 6 shows the equation for the first moment of the velocity or the intensity weighted velocity of the spectral line. ν_i is the velocity of channel corresponding to the flux I_i. The zeroth moment is required to calculate the first moment. Equation 6 gives the mean velocity of the ^{13}CO gas in that voxel.

$$M_2 = \sqrt{\frac{\Delta\nu \sum_{i=0}^{n} I_i (\nu_i - M_1)^2}{M_0}} \tag{7}$$

Equation 7 shows the equation for the second moment of the velocity or the velocity dispersion of the spectral line. It is dependent on the previous two moments, and it gives the width of the spectral line.

The moment maps have been created because they are a good indicator of the kinemtics of the system and will help to describe the dynamics of the clouds. Do clouds from different environments have different preferred direction of spin compared to that of the Galaxy as a whole? This is something that will help to understand if the dynamics of the galaxy aare still dominating at these scales and looked atr different environments may indicate locations where star formation is less previlant due to the dynamics of the galaxy. If the spin of the gas in a cloud was following that of the Milky Way, the spin vector would directly oppose that of the direction of rotation of the galaxy and face towards positive latitude. This is due to the gas in the galaxy rotating differentially, causing shear, meaning the clouds would have a diametricly opposing spin direction. Analysing the velocity of these clouds and calculating theier gradient allows for the direction of the spin vecotr to be derived.

Since the moment 1 map contains the mean velocity for each spatial pixel of the cloud, they will be used to calculate the velocity gradient across the cloud. It is important to note that thes velocities in these moment maps are two-dimensional and are not representative of the cloud as a whole. Additionally, there is no information on the inclination of these clouds, so the velocities have not been corrected for this. The clouds are all at different distances therefore the physical width of each pixel changes drastically over the clouds in the catalogue. This change in physical scale means that any comparison between the kinematics of two clouds would be insignificant. Therefore, as many of the clouds as possible will be brought to a constant physical scale. To do this a gaussian filter was applied to all moment 1 maps. This gaussian filter is a two-dimensional convolution that takes input the standard deviation of the Gaussian used to do the convolution. Figure 8 in Duarte-Cabral et al. 2021 helped to decide at what physical scale to convolve to. As the medial axis length peaks around 5pc, it ws decided that it would be the final physical FWHM beam size after the convolution.

$$\Omega_f{}^2 = \Omega_i{}^2 + \Omega_k{}^2 \tag{8}$$

As the riginal image's beam is described by a Gaussian and the convolution is being completed using a Gasussian function then the resulting image will also be a Gaussian. The variance of the Gaussian can be described by the addition in quadrature of the contributing Gaussian's variances. As the conversion from variance to FWHM is the same for all three factors in the equation, the equation can be written as shown above in Equation 8. Where Ω represents the FWHM beam size, which for the SEDIGISM observations is 28 arcsec (Duarte-Cabral et al. 2021). Working backwards from a final image with a physical beam size of 5pc at the distance of the cloud, the angular FWHM beam size can be calculated. Using the fact that $\sigma = \mathrm{FWHM}/2\sqrt{2\ln 2}$, the standard deviation of the gaussian kernel is calculated. As the original physical beam sizes of all the clouds was less than 5pc, there is the added benefit that the data has been smoothed and so the moment 1 maps now show much better how the velocity changes globally.

The velocity gradient is calculated on a pixel by pixel basis and for an individual pixel it is dependent on the velocities of the surrounding pixels. The gradient in galactic longitude and galactic latitude is calculated individually and is also converted into a magnitude and an angle, each of these are saved as a FITS image. The velocity gradient is defined as the change in velocity per unit distance and these changes are calculated horizontally, vertically and diasgonally. The gradient in the direction of galactic longitude is the sum of the horizontal change in velocity and the horizontal components of the diagonal change in velocity. Typically, all surrounding pixels must be non-zero for the gradient to be calculated, but if the horizontal and vertical suyrrounding pixels are non-zero then they are used exclusively to calculate the velocity graident. The gradient in galactic longitude and latitude is positive for a gradient in the direction of positive galactic longitude and latitude. The angle is in degrees and is zero for positive latitude and increases clockwise towards negative galactic longitude. The methods for calculating the velocity gradient were used on the raw moment 1 maps as well as the many differently smoothed versions and after comparison of several clouds it was obvious that the smoothed moment 1 maps give a much more precise velocity gradient.

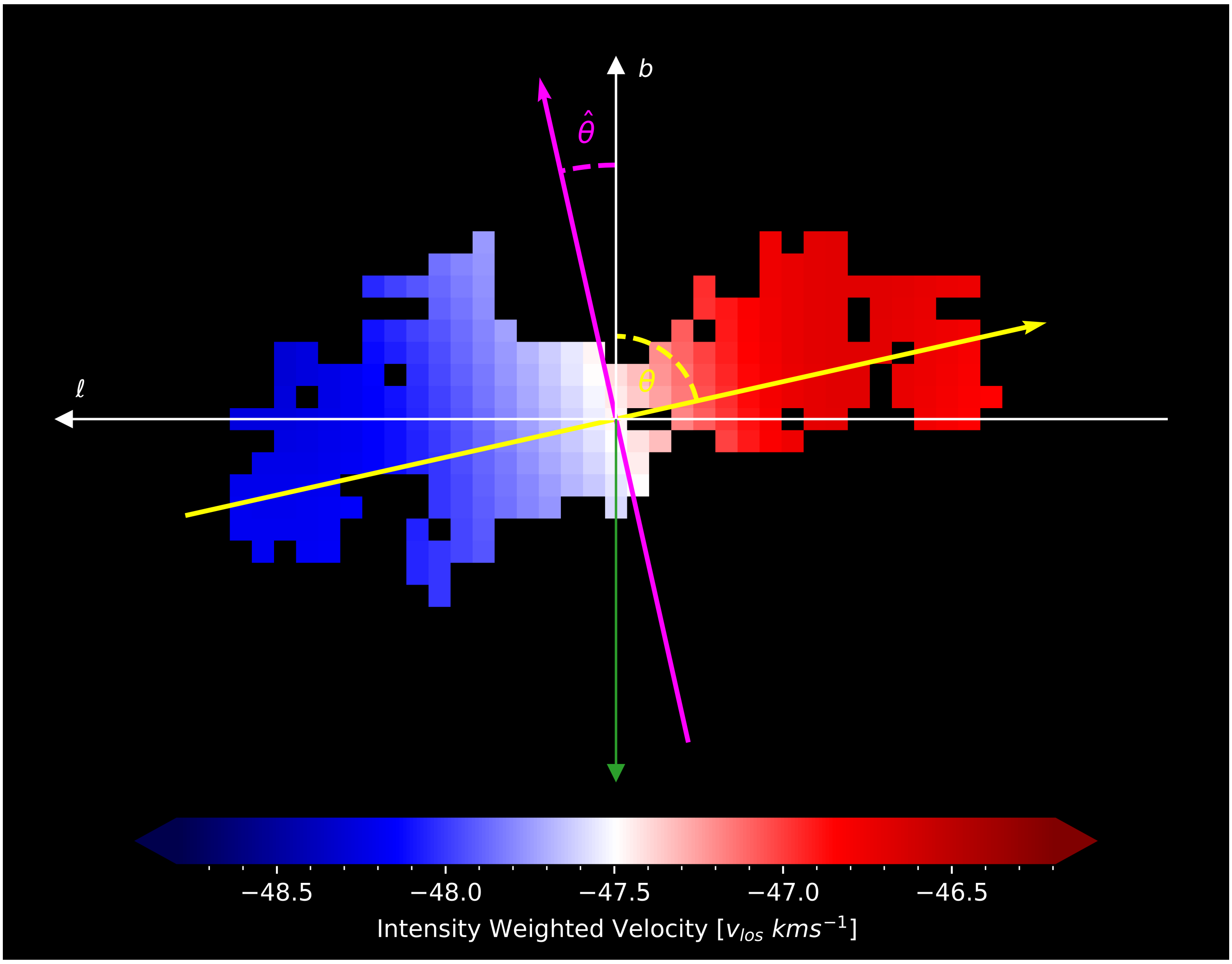

Figure 1: The velocity gradient direction is shown by the yellow arrow and the spin vector is shown by the green arrow. The colourmap behind the gradient is of cloud 4107 from the SEDIGISM catalogue.

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